Chapter 1 Variable

Type	Draft
Status	Not started
Project	My Python Journey

Variables in Python

Let's learn variables step by step, from the basics to interview questions.

1. What is a variable?

A variable is a name used to store a value in memory.

In simple words, a variable is like a container that holds information.

Real-life example:

Imagine you have a box named `age` and you put the number `22` inside it.

In Python:

```
age = 22
```

Here:

- `age` → Variable name
- `=` → Assignment operator
- `22` → Value

VISUAL EXAMPLE

Variable: age

22

Stored value

2. How to create a variable?

In Python, we don't need to declare the data type separately.

Syntax:

```
variable_name = value
```

Examples:

```
name = "Gaythri"age = 22height = 5.4is_student = True
```

Python automatically understands the type of value.

3. Types of variables based on data

Data type	Example	Meaning
String (`str`)	`name = "Gaythri"`	Text
Integer (`int`)	`age = 22`	Whole number
Float (`float`)	`price = 99.5`	Decimal number
Boolean (`bool`)	`is_valid = True`	True or False

To check the data type, use `type()`.

```
age = 22print(type(age))
```

Output:

```
<class 'int'>
```

4. Rules for naming variables

Python has some rules for variable names:

1. A variable name can contain letters, numbers and underscores.
2. It cannot start with a number.
3. Spaces are not allowed.
4. Variable names are case-sensitive.
5. Python keywords cannot be used as variable names.

Valid names	Invalid names
`student`	`2student`
`student_age`	`student age`
`age1`	`student-name`
`_name`	`class`

Remember: `age` and `Age` are two different variables in Python.

5. Assigning multiple variables

Python allows us to assign multiple variables in one line.

Same value to multiple variables:

```
a = b = c = 10print(a, b, c)
```

Output:

```
10 10 10
```

Different values to multiple variables:

```
a, b, c = 10, 20, 30print(a, b, c)
```

Output:

```
10 20 30
```

6. Changing the value of a variable

You can change a variable's value whenever needed.

```
age = 20age = 25print(age)
```

Output:

```
25
```

The latest assigned value is printed.

7. Practice time

Try these three questions. Don't run the code yet; predict the output.

Variable practice

1 / 3

What is the output?

x = 10x = 20print(x)

10

20

Error

PreviousCheck answer